AI Museum of the Future

Technical reference: models, hyperparameters, metrics
and end-to-end path to the museum exhibition

Team project — ESPRIT / Deep Learning
Document for oral defence (professor)

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Abstract

This note lists every major model used (or planned) in the project, the hyperparameters that control retrieval and agents, the measured metrics, and the full pipeline from archival documents to a curated museum exhibition. **Deployed today:** text LLM (Groq), local embeddings (MiniLM), vision understanding (Qwen). **Image generation** is not in production; we document **Flux.1** as the technical candidate for optional hall illustrations.

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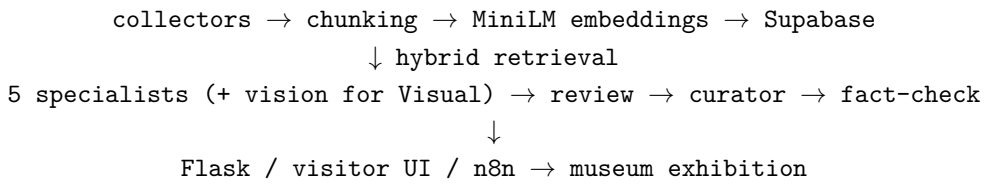
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1 End-to-end path: from request to museum

The system turns a visitor (or curator) request into a multi-section exhibition grounded in documentary evidence.

1. **Input.** Country / year range / topic (default Tunisia, 2020–2026), via Flask dashboard, visitor UI, or n8n webhook.
2. **Knowledge base.** Documents collected, chunked, embedded, stored in Supabase (Postgres + pgvector + full-text search).
3. **Hybrid RAG.** For each specialist: vector search + keyword search, fused, filtered, diversified; optional LLM rerank (off by default).
4. **Five specialists (parallel).** Historian, Sociologist, Technology, Culture & Art, Visual — each with its own search queries and prompt.
5. **Multimodal (Visual).** Up to N images are described by a vision model; descriptions become evidence lines `vision:...`.
6. **Governance.** Peer reviewer → Curator (4–6 sections) → Fact-checker (unsupported claims trigger one curator rewrite).
7. **Persistence.** Run saved as `runs/<run_id>/run.json` (prompts, evidence, insights, timings).
8. **Experience layer.** Exposition page, visitor 3D halls (corpus `image_url` on walls), optional n8n delivery / Binary image nodes.



2 Models catalogue

2.1 Models deployed in the running system

Role	Model / component	Provider	Env / notes
Text generation & reasoning (all specialists + reviewer + curator + fact-checker by default)	`openai/gpt-oss-120b`	Groq (OpenAI-compatible API)	`LLM_MODEL`; `LLM_BASE_URL=https://api.groq.com/open`
Optional stronger curator	same or override	Groq / other	`CURATOR_MODEL`
Vision / multimodal **understanding**	`qwen/qwen3.8-27b`	Groq (vision-capable)	`VISION_MODEL`; describes real corpus images
Optional retrieval rerank	defaults to `LLM_MODEL`	same	`RERANK_MODEL`; off by default in agent searches
Embeddings (local)	`paraphrase-multilingual-MiniLM-L12-v2`	FastEmbed (on-device)	384 dimensions; ∼120 MB download once; no API key

Role	Model / component	Provider	Env / notes
Vector DB	pgvector in Supabase	Supabase	`vector(384)`; RPC `match_chunks` + FTS
Orchestration	n8n workflow	self-hosted / cloud	start → poll → summary / museum URLs
Application	Flask + static frontends	Python	visitor, dashboard, Agent lab, exposition API

2.2 Image generation candidate (not deployed): Flux.1

- **Name:** Flux.1 (Black Forest Labs) — modern text-to-image model.
- **Role in our architecture (planned / oral answer):** optional *illustration* of museum halls or atmospheric panels when no corpus photo exists — always labelled as generated, never mixed silently with documentary evidence.
- **Why Flux.1 is a good technical answer:** strong prompt adherence and image quality vs older Stable Diffusion baselines; available via API or GPU inference; industry-standard choice in 2025–2026 demos.
- **How Flux.1 works (high level):**
  1. Encode the text prompt (and optional style constraints).
  2. Sample from noise in a latent space.
  3. Iteratively transform / denoise (flow-matching / diffusion-family dynamics) toward an image matching the prompt.
  4. Decode the latent to a pixel image.
- **Why we do not use it in production today:** free-tier credits, documentary integrity (museum of memory prefers real `image_url`), and multimodal priority = *understanding* captions vs pixels.

One-line oral answer: "For generation we would use Flux.1; for understanding we already use Qwen vision; today the walls show real archive images."

3 What each agent does (same LLM, different prompts)

All specialists share `agents/base_agent.py`: gather evidence → LLM call (JSON) → citation filter → insight or abstain.

Agent	Model	Focus / special knobs
Historian	`LLM_MODEL`	Timelines, events, long memory; may write in French.
Sociologist	`LLM_MODEL`	Society, lifestyles; own search + **HyDE** (hypothetical document expansion, cached under `HYDE_CACHE_DIR`).
Technology	`LLM_MODEL`	Innovations, tools, implications; dedicated search queries & prompt v5.
Culture & Art	`LLM_MODEL`	Arts, media, symbols; category-aware search.
Visual	`LLM_MODEL` + `VISION_MODEL`	Images first: describe up to `VISUAL_MAX_IMAGES` (default 3), then interpret. Search category `image`.

Agent	Model	Focus / special knobs
Peer reviewer	`LLM_MODEL`	Agreements, tensions, weak findings across insights.
Curator	`LLM_MODEL` or `CURATOR_MODEL`	Title, concept, 4–6 sections, artifacts from specialists only; one retry with feedback on validation failure.
Fact-checker	`LLM_MODEL`	Narrative vs cited evidence; unsupported → one curator rewrite.

Hard citation rule: every finding must contain at least one `[chunk_id]` present in that agent's evidence; uncited findings are dropped; if none remain, the agent abstains and appears in `missing_perspectives`.

4 Hyperparameters and configuration knobs

4.1 LLM / runtime

Parameter	Default	Effect
`LLM_MODEL`	`openai/gpt-oss-120b`	Text reasoning / generation
`VISION_MODEL`	`qwen/qwen3.8-27b`	Image understanding
`CURATOR_MODEL`	(unset = LLM)	Optional stronger curator
`LLM_MAX_CONCURRENCY`	3	Parallel LLM calls (TPM-friendly)
`LLM_ATTEMPTS`	4	Retries with provider-aware wait
Response format	JSON object	Structured agent outputs
Specialist timeout	120 s (often 400 s free tier)	Pipeline `-timeout`
Full run LLM calls	∼15–20	Quota planning on Groq free tier

4.2 Embeddings & chunking

Parameter	Default / choice	Rationale
Embedding model	MiniLM multilingual L12	Multilingual FR/EN/AR corpus
Dimension	384	Matches `vector(384)` schema
Effective window	128 tokens	Model ignores tokens beyond 128
Chunking strategy	Sentence packing to 128 + title prefix	Chosen over fixed-512 (57% text unse
Chunking MRR@10 (selection)	0.47	Passage-level offline comparison

4.3 Hybrid retrieval (`SearchConfig`)

Parameter	Default	Meaning
`candidate_pool`	40	Candidates per channel before fuse/filter
`statistics_weight`	0.35	Down-weight World Bank-style stats rows
`max_statistics`	1	At most one stats row in a result set
`duplicate_threshold`	0.6	Token Jaccard near-duplicate cut
`rerank_pool`	20	Candidates sent to LLM reranker
`relax_country`	true	Top-up outside country if too few hits
`rerank_agent_searches`	false	LLM rerank off for agent own-search
Default year filter	2020–2026	Exhibition default window
`keep_undated`	true	Many docs lack `published_at`
Result diversity	1 chunk / document	Avoid same-doc flooding

4.4 Multimodal

Parameter	Default	Meaning
`VISUAL_MAX_IMAGES`	3	Vision LLM calls per Visual run
Download User-Agent	required	Wikimedia returns 403 without it
Cache	`multimodal/cache/`	Avoid re-describing same URL
Output	structured JSON + text line	Fed as `vision:` evidence; optional DB `vision_*`

4.5 Curator validation

- Section count must be in $[4, 6]$.
- Non-empty title / concept / narratives.
- Artifacts must come from specialists (no invented items).
- One retry with explicit feedback; then `CuratorValidationError`.

4.6 Flux.1 (if enabled later) — typical generation knobs

Knob	Typical range	Role
Prompt (+ negative)	free text	Subject, style, documentary vs illustrative
Steps / guidance	model-dependent	Quality vs latency / cost
Resolution	e.g. 1024×1024	Hall panel size
Seed	fixed for demos	Reproducibility
"AI illustration"	Ethics / museum honesty	

5 Metrics and evaluation results

5.1 Retrieval (documented in `evaluation/`)

Test set: **16 queries**, $k = 5$, term-rule relevance proxy (`retrieval_cases.json`).

Run 2026-10-14 (README table — often cited in pitch).

Method	Precision@5	Hit@5	MRR	Stats share
Vector only	0.41	0.75	0.52	0.11
Hybrid	0.68	0.94	0.83	0.00
Hybrid + LLM rerank	**0.72**	**1.00**	**0.92**	0.00

Update 2026-10-15 (full reranks completed). Vector 0.50/0.88/0.61; hybrid 0.70/1.00/0.94; hybrid+rerank 0.56/1.00/0.75. With every rerank executed, reranking **hurt** precision — hence **rerank off by default**.

Metric definitions.

- **Precision@k:** fraction of top-k results that match relevance rules.
- **Hit@k:** fraction of queries with ≥ 1 relevant result in top-k.
- **MRR:** mean of 1/rank of the first relevant result.

Caveat: Tunisia-specific precision stays low (∼0.2–0.4): few Tunisia chunks (∼75, many statistics). Better retrieval cannot invent missing archives.

5.2 Chunking selection

Offline comparison on long documents / passage questions (EN/FR/AR): sentence packing to the embedder's 128-token window + title prefix selected (MRR@10 ≈ 0.47 in D1 draft notes).

5.3 Exhibition run metrics (`evaluate_run`)

Metric	Meaning
coverage	Specialists with cited findings / 5
abstained / errors	No valid citations / provider timeout
findings_dropped_uncited	Guardrail: uncited claims removed
citation_survival	kept / (kept + dropped)
own_search_results_per_agent	Evidence each agent retrieved itself
review_*	Peer-review agreements, tensions, weak findings
fact_check_supported	Sections fully supported by cited text
faithfulness (`-judge`)	LLM judge: finding supported by its citations
llm_calls / timings_s	Cost and latency of the run

5.4 Multimodal qualitative metric

Caption "Protest in Mexico City" vs vision: beach/road scene, no protest → `caption_mismatch`. Demonstrates why vision understanding matters beyond text RAG.

6 Multimodal module — image understanding (not generation)

1. Read `image_url` from document metadata.
2. Download image (browser User-Agent).
3. Call `VISION_MODEL` → structured JSON description.
4. Disk-cache; attach `vision:` line to Visual evidence.
5. Optionally write `vision_*` into Supabase so hybrid search can match visual content, not only captions.

Exposition UI can show "what the AI sees" via `POST /api/exhibition/<run_id>/vision`.

7 Orchestration and museum delivery

- **Flask API:** start / cancel / poll / list runs; vision describe; Agent lab `POST /api/agents/<name>/analyze`.
- **n8n:** `Exhibition_pipeline_end_to_end.json` — webhook/schedule → start pipeline → poll (∼60 s) → summary + museum / halls URLs; Binary download of corpus images for walls.
- **Visitor UX:** warm public frontend + 3D rooms (specialist galleries) + immersive visit; login for protected routes (demo `visiteur` / `musee2126`).
- **Dashboard:** Agent lab (test each specialist), exhibition board, data browser.

8 Trust, security, and limits

- Traceability via mandatory `[chunk_id]` citations.
- Structured JSON + curator / fact-check validators.
- Secrets in `.env` (never committed); optional Bearer tokens for APIs.
- Free-tier TPM can block curator rewrite after fact-check.
- Citation $\neq$ proof: relevance and fact-check remain necessary.
- Image generation (Flux.1) kept optional to protect documentary honesty.

9 Quick oral cheat-sheet

1. **Text brain:** Groq `gpt-oss-120b`.
2. **Memory:** MiniLM 384-d + Supabase hybrid RAG.
3. **Eyes:** Qwen vision — understand photos.
4. **Paintbrush (future):** Flux.1 — generate illustrations only.
5. **Best retrieval numbers to quote carefully:** hybrid+rerank 0.72 / 1.00 / 0.92 (14 Oct) *or* hybrid 0.70 / 1.00 / 0.94 with rerank off (15 Oct finding).
6. **Museum result:** 4–6 curated sections + corpus images on walls, not AI-invented history.

Sources in repository: `README.md`, `agents/llm_config.py`, `rag/vector_store/hybrid_retriever.py`, `multimodal/READM` `evaluation/README.md`, `docs/REPORT_D1.md`. Confirm live `.env` model names before the defence if the team changed providers.